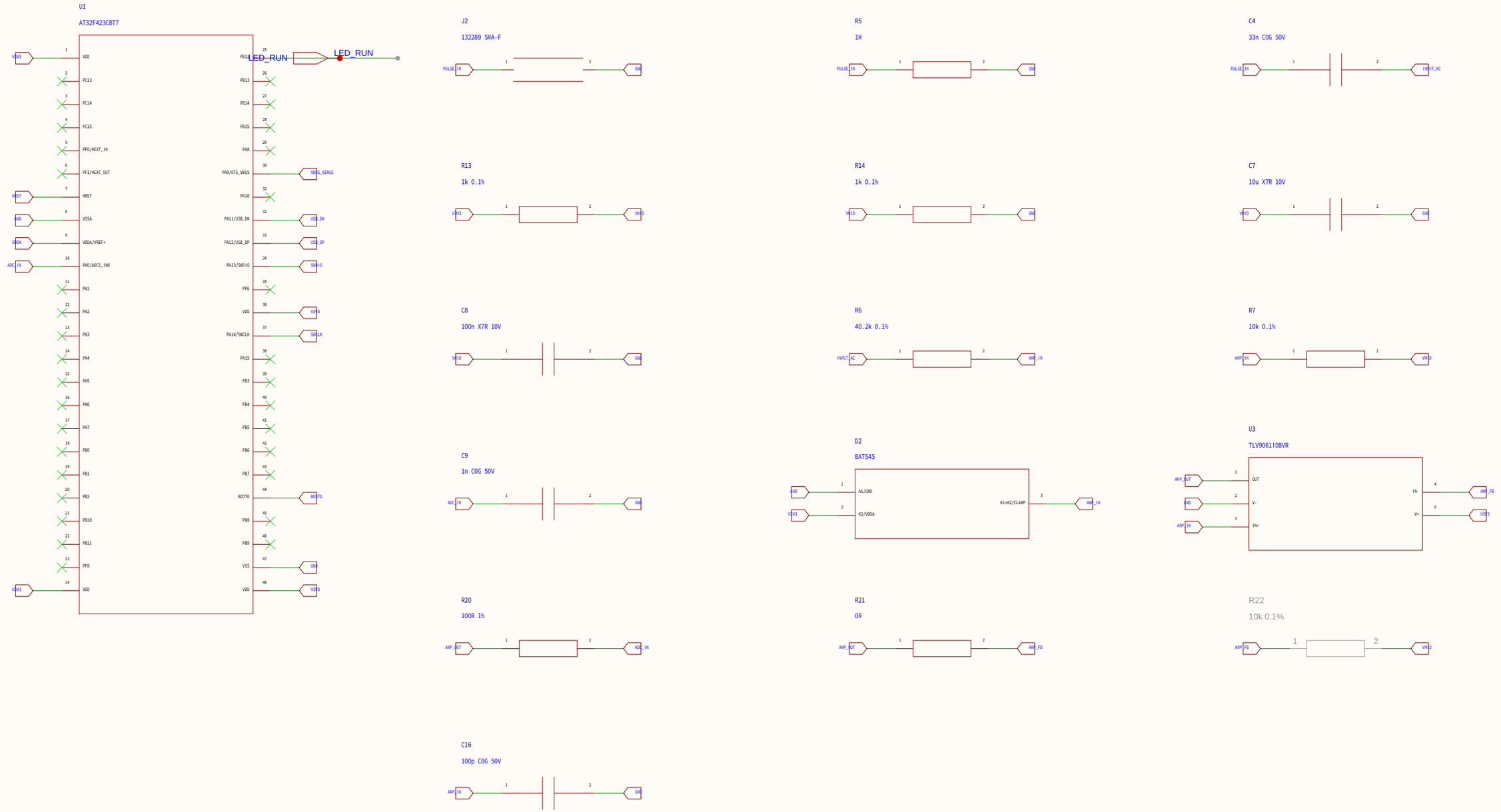
Aster-MCA-Nano-AT32-v0.3-USB-A-LED

01 MCU and analog pulse input | ENGINEERING PROTOTYPE - performance unverified

SMA -> C4/R6/R7/C16 -> U3 TLV9061 -> R20/C9 -> PA0 ADC. U3 uses DBV pinout, powered from 3.3V.

Default gain ~0.199; input load ~50k. R21=0R FITTED; R22=10k DNP. Optional gain ~1: R21=40.2k + fit R22.

C4=33nF C0G. U3 decoupling C17=100nF. ADC isolation R20=100R, C9=1nF. No complete MCA firmware supplied.



Aster-MCA-Nano-AT32-v0.3-USB-A-LED

02 USB-A, power and boot | ENGINEERING PROTOTYPE - performance unverified

USB DFU: hold BOOT on power-up. Factory nBOOT1=1. HICK USB clock requires ACC / SOF calibration.

ADC erratum ES0010: set ADC_MSIC.XTEST=0x09 after ADC clock ON, before enabling ADC/calibration.

D1
USBLC6-2SC6



U2
XC6206P332MR-G



C1
1u X7R 10V



C2
4.7u X7R 10V



C3
100n X7R 10V



SW1
B3U-1000P BOOT



R17
10k



R16
10k



R18
10k



R19
0R



C10
100n X7R 10V



C11
100n X7R 10V



C12
100n X7R 10V



C13
100n X7R 10V



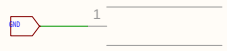
C14
1u X7R 10V



C15
100n X7R 10V



TP1
GND



TP2
V3V3



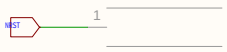
TP3
SWDIO



TP4
SWCLK



TP5
NRST



TP6
VMID



TP7
VDDA



TP8
ADC_IN



J1
1001-011-01101

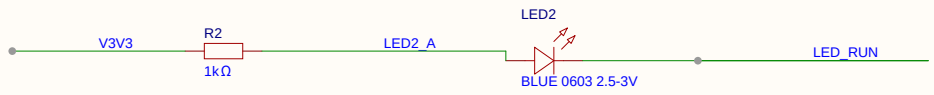


C17
100n X7R 10V





PWR: 3.3V rail



RUN: active low, 1Hz heartbeat